Penile doppler ultrasound in patients with erectile dysfunction (ED): role of peak systolic velocity measured in the flaccid state in predicting ar...

INTRODUCTION: The use of the penile peak systolic velocity (PSV) measured in the flaccid state during penile color Doppler ultrasound (PCDU) examination has been questioned without substantial evidence. AIM: To assess the validity of PSV measured in the flaccid state during PCDU, in patients consulting for erectile dysfunction (ED). METHODS: A consecutive series of 1,346 (mean age 55.0 +/- 12.0 years) male patients was studied. MAIN OUTCOMES MEASURES: All patients underwent PCDU performed both in the flaccid state and dynamic (after prostaglandin E1 stimulation) conditions. A subset of 20 subjects with uncomplicated type 2 diabetes underwent diagnostic testing for silent coronary heart disease by means of adenosine stress myocardial perfusion scintigraphy (SPECT). In these subjects penile arterial flow was simultaneously assessed by PCDU before and after systemic adenosine administration. RESULTS: Flaccid PSV showed a significant ($r = 0.513$, $P < 0.0001$) correlation with dynamic PSV. Receiver operating characteristic (ROC) curve analysis demonstrated that when a threshold of 13 cm/seconds was chosen, flaccid PSV was predictive for dynamic PSV < 25 and <35 cm/seconds with an accuracy of 89% and 82%, respectively. Among the subset of patients who underwent SPECT, an impaired coronary flow reserve (ICFR) occurred in nine cases (45%). When the same threshold of <13 cm/seconds was chosen, PSV before SPECT was predictive of ICFR with an accuracy of 80% (area under the ROC curve = 0.798 +/- 0.10; $P < 0.05$). After adjustment for confounders, anxiety symptoms were related to dynamic PSV (Adj. $r = -0.154$, $P < 0.05$) but not to flaccid PSV. CONCLUSIONS: Our results show that flow in the cavernosal arteries can be routinely evaluated by PCDU in the flaccid state. Performing PCDU only in the flaccid state allows identifying subjects with pathological dynamic PSV with accuracy higher than 80%. Furthermore, our preliminary data suggest that the same examination could identify diabetic subjects with ICFR with an accuracy of 80%.